How equal are tests of FAIRness? - A comparative evaluation from a domain-specific perspective

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The FAIR principles (Wilkinson et al. 2016) are an important guideline for sustainable research data management (RDM). They take data as well as metadata into account and define criteria for these to be Findable, Accessible, Interoperable and Reusable. Thereby most of the criteria focus on technical aspects, such as persistent identifier (principle F1), retrieval communications protocol (principle A1), qualified references (principle I3) and accessible data usage license (R1.1), and therefore seem to be eligible for automatic verification. A few of the criteria come with open instantiation details, such as searchable resource (F4), detailed provenance (R1.2) and domain-relevant community standards (R1.3). In addition, domain-specific interpretations of FAIR have been argued for (Helling et al. 2022).

Thus, to evaluate the FAIRness of a specific resource, there are several automatic and non-automatic approaches, including automatically evaluated test collections, survey tools and lists (Lang et al. 2023). The aim of this work is to compare the scoring of automatically evaluated test collection in a real-world, domain-specific context: data sets from computational literary studies.

Our study analyzes output from three tools in particular. These selection reflects choices made in previous investigations (Kontsioti 2023, Lang et al. 2023).

- F-UJI[1], from the FAIRsFAIR project "Fostering FAIR Data Practices in Europe". We apply the tool in version 3.2.0, with the FAIRsFAIR Data Object Assessment Metrics v0.5 (Devaraju et al. 2022) and DataCite support enabled.
- FAIR-Checker[2] (Gaignard et al. 2023) from the Bioinformatics domain. We apply the tool in version v1.3.
- FAIR Enough[3], developed by the Institute of Data Science at Maastricht University, which allows to switch between several test collections. For this study, we apply the FAIR Enough metadata maturity indicators, the FAIR Enough data maturity indicators and the FAIR Evaluator maturity indicators, the latter based on the collection in the FAIR Evaluation Services[4].

As input, all tools take a persistent identifier (e.g. DOI) or URL of a resource to assess. Next to the results of every test conducted, each tool outputs an overall FAIRness score.

We use an existing set of resources, with different FAIR levels to be expected, and compare the outputs of the tools with respect to their FAIRness assessment. The resources are non-paper research outputs (NPROs, Helling et al. 2024) from the SPP 2207, a priority programme on Computational Literary Studies including several projects from universities in Germany and Switzerland. Their DOIs and URLs have been taken from the Zenodo community "SPP 2207 Computational Literary Studies"[5], including the resource types “Dataset”, “Software” and “Other”; from the programme's web page[6], including the resource types "Datasets and analyses" and "Tools and platforms" and a Zotero library[7] including the resource types: “Software”, “Dataset” and “Web Page”. Reading-based resources, such as annotation guidelines, questionnaires, tutorials and blogs have been excluded.

Since the focus of this work is on the comparison of the tools, we included resources from several levels of expected FAIRness, from persistent datasets in Zenodo, to short-term online data demos without persistent identifier.

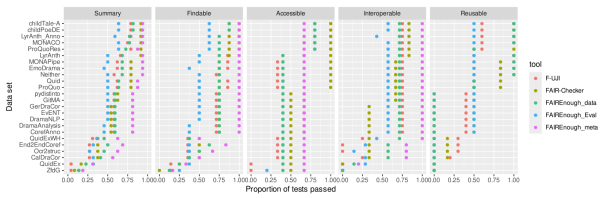

Figure 1: Performance of the example data sets on different FAIRness evaluation tools. For Reusability, FAIREnough_meta score is equal to FAIREnough_data.

Results and Discussion

Looking at the proportion of individual tests a data set passed reveals considerable variability between tools. The smallest differences between tools are seen the summary FAIRness scores that include all the tests conducted (Fig. 1). The FAIR Enough metadata maturity indicators tool, while assigning the same score to all data sets in the "Accessible" category, accounts for the majority of maximum values in the "Findable" and "Interoperable" categories as well as in the summary score. As this tool, opposed to the others, specifically evaluates only the metadata of the data set it might be considered as a special case. If not considering it in the comparison, differences between tools are less pronounced.

F-UJI, FAIR-Checker and FAIR Enough data maturity indicators often yield very similar results; deviations in individual categories can sometimes be attributed to the small number of tests a tool applies in a certain category.

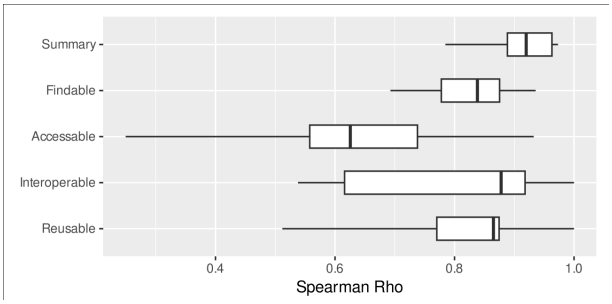

Figure 2: Pair-wise rank correlation coefficients for tool comparisons.

Ranking the data sets by FAIRness relative to each other the different tools produce comparable results. Pair-wise rank correlation analysis shows positive correlation between the results from all tools. Especially in the summary scores, correlation coefficients are consistently high ranging from 0.78 to 0.97 (ignoring the metadata-based tool for the reasons outlined above would even push the minimum to 0.88). Correlation coefficients are lower when looking into individual categories (Fig. 2).

Our observations indicate that automatically evaluated test collections are a valid option to evaluate FAIRness in data sets from computational literary studies. The different tools overall produce roughly comparable scores and rankings. Especially if only a rough overall estimate is required, F-UJI, FAIR-Checker and FAIR Enough data maturity indicators provide very similar results. However, when looking into the individual categories, there is variability in the results due to the different metrics (i.e. test collections) applied by the different tools. Looking e.g. at the DramaAnalysis data set, summary estimates are - except for meta data scores - very consistent, but detailed assessments of findability and reusability differ considerably between tools. Where detailed results matter, it makes for a valid recommendation to look into these tests in detail and, if necessary, apply multiple tools to gain a combined result tailored to domain-specific needs.

Notes

1. https://www.f-uji.net/
2. https://fair-checker.france-bioinformatique.fr/
3. https://fair-enough.semanticscience.org/
4. https://fairsharing.github.io/FAIR-Evaluator-FrontEnd/#!/
5. https://zenodo.org/communities/spp_cls/records
6. https://dfg-spp-cls.github.io/achievements.html
7. https://www.zotero.org/groups/4667571/spp-cls-achievements

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